

# Epidemiology of Chronic Kidney Disease and Dialysis

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The global average life expectancy is now 73.4 years—an enormous increase from 61.7 years in 1980, due in part to declining mortality from communicable, maternal, neonatal, and nutritional disease.<sup>1,2</sup> On the other hand, with the aging of the population, chronic kidney disease (CKD) has become one of the most common noncommunicable diseases in the world, as well as a leading cause of mortality. Half of the people in the United States are expected to develop CKD during their lifetime.<sup>3</sup> Associated with huge health system costs, particularly with kidney failure, CKD has come into focus as a common, morbid, and often preventable disease.<sup>4</sup>

## DEFINING CHRONIC KIDNEY DISEASE

### Chronic Kidney Disease Staging

Over the past decade, the definition of CKD has evolved to incorporate advances in knowledge about prognosis.<sup>5</sup> The National Kidney Foundation's Kidney Disease Outcomes Quality Initiative guidelines published in 2002 classified CKD into five stages on the basis of glomerular filtration rate (GFR) and signs of kidney damage (pathologic abnormalities or markers of damage in blood, urine, or imaging studies).<sup>6</sup> This staging system was modified in the 2012 Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guidelines to reflect the independent contributions of GFR, albuminuria, and cause of CKD.<sup>7</sup> The two-dimensional "heat map" that classifies CKD in GFR categories (G stages) and albuminuria categories (A stages) has been widely accepted and robustly validated, but less is known about the third dimension, as routine reporting on the cause of CKD is relatively uncommon (Fig. 80.1).<sup>8</sup>

### Classification Based on Estimated Glomerular Filtration Rate

Serum creatinine is the most commonly used filtration marker for the estimation of GFR, and GFR estimating equations incorporating serum creatinine, as well as age and sex, generate precise and accurate assessments of kidney function in the vast majority of both CKD and non-CKD populations.<sup>9,9a</sup> In North America, the best available GFR estimating equation is that developed by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI).<sup>10</sup> The accuracy of GFR estimation also has benefited from universal standardization of the creatinine laboratory assay (see Chapter 3).<sup>11</sup> Serum cystatin C is an additional filtration marker used in the estimation of GFR, with an

equation that uses both cystatin C and serum creatinine recognized as being the most accurate.<sup>9a</sup>

### Classification Based on Albuminuria

Multiple techniques exist to estimate urine albumin excretion. The semiquantitative dipstick method is most common, followed by spot and 24-hour urine protein or albumin excretion assessments. All are subject to limitations. Dipstick testing is inexpensive, but the diagnostic accuracy can be low,<sup>12,13</sup> particularly for moderately increased levels of albuminuria. Although a full day's collection of urine albumin is the gold standard for albuminuria estimation, the process is tedious and imposes substantial burden on the patient. Thus, the KDIGO guideline recommends a spot collection of urine, with early morning timing preferred, as the initial screening test.<sup>7</sup> Urine albumin is recommended over urine protein assessment given the superiority of the laboratory assay and standardization, as well as the relative importance in prognosis. Spot samples are standardized to urine creatinine to normalize to 24-hour excretion and account for any differences in concentration (e.g., albumin-to-creatinine ratio [ACR]). Abnormalities in GFR or albuminuria must persist for more than 3 months to meet the definition of CKD (see Chapter 3).

### Classification Based on Chronic Kidney Disease Cause

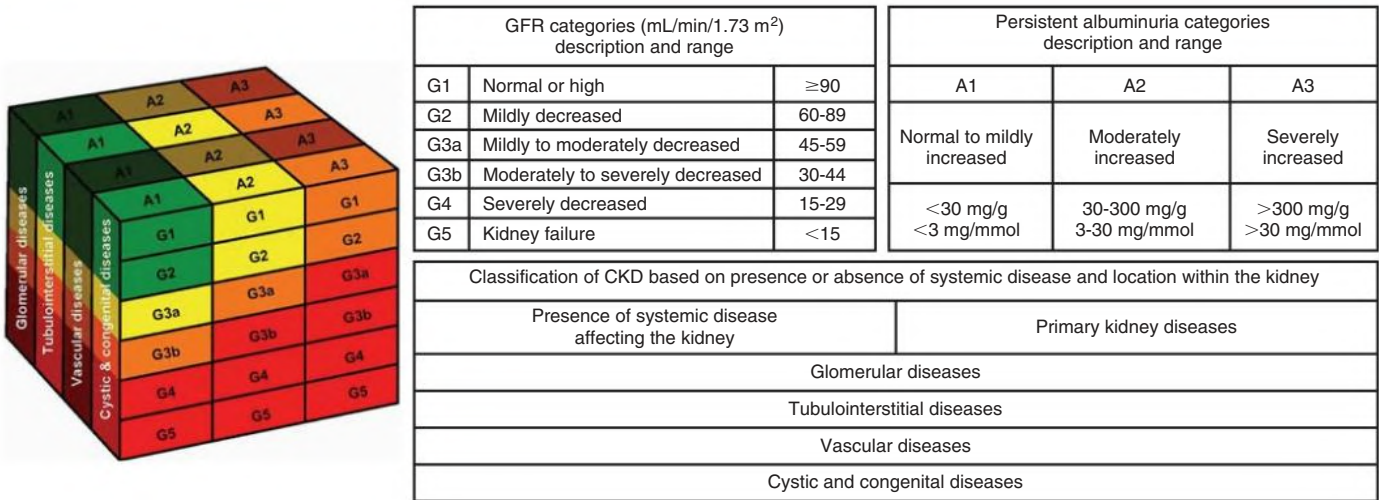
The cause of kidney disease (the "C" in the KDIGO CGA staging) is the least studied aspect of the CKD definition. The proposed KDIGO classification categorizes disease by the presence of underlying systemic disease and the location of the abnormality (see Fig. 80.1). For example, diabetic nephropathy might be classified as a systemic disease affecting the glomerulus, and obstructive nephropathy as a primary kidney disease affecting the tubulointerstitium. Limitations of this approach include the infrequent use of kidney biopsy, the gold standard for assessment of pathologic location of kidney abnormalities. On the other hand, decisions not to perform kidney biopsies are often driven by the uncertain benefit of precise localization of pathologic abnormalities in terms of prognostic implications—a vicious circle begetting sparse evidence.

### Rationale for the Chronic Kidney Disease Definition

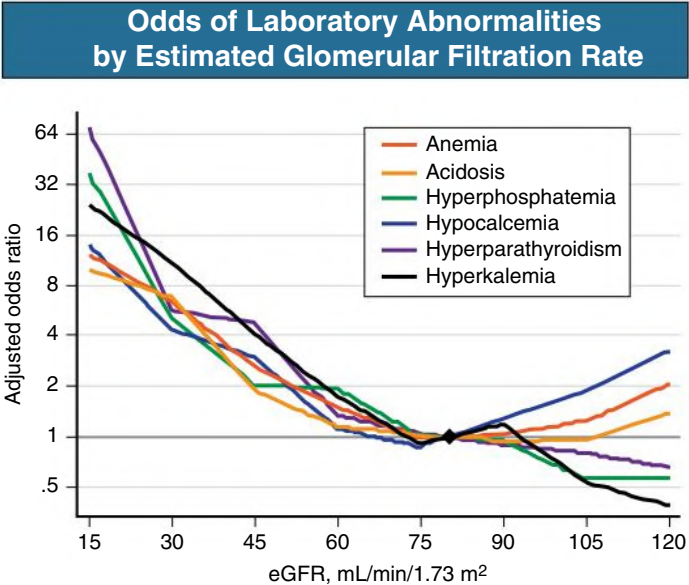
#### Associations With Adverse Outcomes

The cut-points of GFR and albuminuria used to define CKD (<60 mL/min/1.73 m<sup>2</sup> and ≥30 mg/g, respectively) have been rigorously

Chronic Kidney Disease Classification According to Glomerular Filtration Rate and Albuminuria



**Fig. 80.1** Depiction of Chronic Kidney Disease (CKD) Classification: G Stages and A Stages. The cube denotes how the three components of the CKD classification scheme (glomerular filtration rate [G], albuminuria [A], and underlying kidney disease) interact to influence the risk of progression to kidney failure. This risk is represented in qualitative terms (lowest to highest) by *green, yellow, orange, and red* colors.



**Fig. 80.2** Odds of laboratory abnormalities according to estimated glomerular filtration rate (eGFR) among people from the general US population. (From the continuous National Health and Nutrition Examination Survey [1999–2010].)

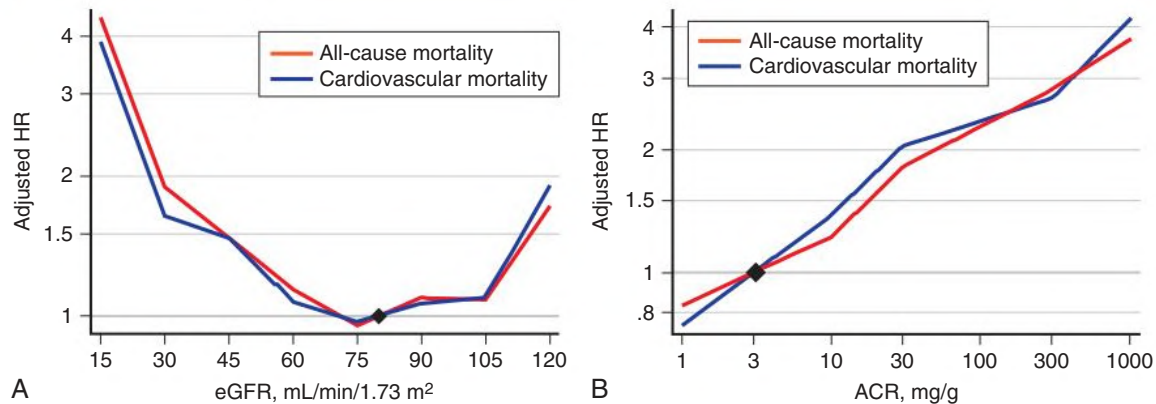
examined. Each represents a higher risk state compared with normal levels for a variety of outcomes. For example, the relationship between GFR and the prevalence of anemia, acidosis, hyperphosphatemia, hypoalbuminemia, and hyperparathyroidism demonstrates a continuous increase in risk with GFR less than 60 mL/min/1.73 m<sup>2</sup> (Fig. 80.2).<sup>14</sup> Furthermore, the risk for future kidney outcomes, including acute kidney injury (AKI), CKD progression, and kidney failure, all exhibit graded associations with lower GFR.<sup>15</sup> Although relationships with progression to kidney failure with kidney replacement therapy (KRT) are strongest, both lower eGFR and more severe albuminuria are independently associated with cardiovascular mortality and

all-cause mortality as well (Fig. 80.3).<sup>8</sup> For example, for a given value of GFR, more severe albuminuria is associated with higher risk for adverse events compared with albuminuria levels less than 30 mg/g.<sup>4</sup>

Persistent Relationships Within Subgroups

Although there remains some controversy about the use of different thresholds for different groups of people,<sup>16</sup> the risk relationships between eGFR and ACR with adverse outcomes are remarkably robust in different subgroups, including older and younger adults and men and women. For example, while reduced GFR is more common among older individuals compared to younger individuals, it remains a strong

### Hazard Ratio of Adverse Outcomes Associated With Estimated Glomerular Filtration Rate and Albuminuria



**Fig. 80.3** Hazard ratios (HRs) of adverse outcomes associated with estimated glomerular filtration rate (eGFR) (A) and albuminuria (B) among people from the general US population. ACR, Albumin-to-creatinine ratio. (From the National Health and Nutrition Examination Survey III [1988–1994] and the continuous National Health and Nutrition Examination Survey [1999–2010].)

risk factor for KRT compared with higher levels of GFR.<sup>17</sup> Similarly, the risk relationships between ACR and adverse outcomes appear fairly consistent in women compared with men, counter to the proposed sex-specific definitions that assign a higher ACR threshold in women (Fig. 80.4).<sup>18</sup>

## GLOBAL BURDEN OF CHRONIC KIDNEY DISEASE

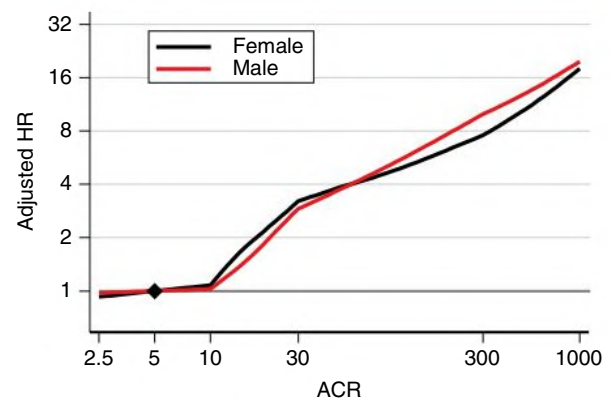
Accurate estimations of CKD prevalence are difficult because tests for filtration markers and albuminuria are not uniformly performed; however, many believe the global burden of CKD to be large and increasing.<sup>19</sup> The best estimates are derived from nationally representative samples, such as the National Health and Nutrition Examination Survey (NHANES) in the United States; estimates based on populations receiving health care may overestimate (or, in some cases, underestimate) disease prevalence.<sup>20</sup> In the United States, estimates of CKD prevalence based on a one-time measurement of serum creatinine recently placed the proportion of the population with stage G3 or G4 CKD at 6.9%; using a definition of ACR greater than 30 mg/g or eGFR less than 60 mL/min/1.73 m<sup>2</sup>, the prevalence of any CKD was 14.2%.<sup>21</sup> This definition likely overestimates prevalence because the persistence of albuminuria was not taken into account. Publications from countries in Europe show wide variation in CKD prevalence, from 3.3% in Norway to 17.3% in northwest Germany, but this may reflect differences in sampling strategy rather than true population-level differences.<sup>22</sup> Estimates in lower-income countries are less available, but most place the global population burden between 8% and 16%, affecting nearly 500 million adults worldwide in 2010.<sup>23</sup>

## RISK FACTORS FOR CHRONIC KIDNEY DISEASE

### Age

Many risk factors have been implicated in the development of CKD. Age is a strong risk factor for GFR less than 60 mL/min/1.73 m<sup>2</sup>, and so population aging is almost inevitably associated with an increase in CKD burden. Interestingly, although incidence rates of KRT tend to be highest among persons older than 65 years, older age is often associated with a lower risk for developing KRT in multivariable-adjusted analyses.<sup>4,24</sup> In other words, for the same level of kidney function, a

### Hazard Ratio of Adverse Outcomes Associated With Estimated Glomerular Filtration Rate, by Sex



**Fig. 80.4** Hazard ratios (HR) of adverse outcomes associated with albuminuria, by sex. ACR, Albumin-to-creatinine ratio. (From Nitsch D, Grams M, Sang Y, et al. Associations of estimated glomerular filtration rate and albuminuria with mortality and renal failure by sex: a meta-analysis. *BMJ*. 2013;346:f324.)

younger person is at higher risk for progression to end-stage kidney disease (ESKD) than an older person. This observation may be due in part to the fact that mortality is lower in younger persons and that younger persons may be more likely to accept or be prescribed a replacement therapy (dialysis or transplantation) compared with older persons.

### Sex and Race/Ethnicity

Risk for CKD differs by other demographic factors as well. Women are generally at higher risk for incident CKD but lower risk for incident KRT than men.<sup>2</sup> In the United States, Black individuals have a higher prevalence of KRT than White individuals. Much of the difference in CKD burden stems from long-standing and pervasive structural racism resulting in differences in socioeconomic status and access to health care.

Differences in the frequency of specific genetic variants also partially underlie this disparity. Sickle cell trait and the *APOL1* high-risk genotype, present in 7% and 13% of persons of African descent, respectively, confer higher risk for CKD and KRT.<sup>25,26</sup> The presence of sickle cell trait may confer risk via subclinical sickling in the oxygen-poor medulla, leading to impairments in concentrating ability and hematuria over time. The mechanism of the *APOL1* high-risk genotype is as yet uncertain, but it appears to be mediated through the development of albuminuria. Other racial/ethnic groups also have higher risk of KRT, including persons of Hispanic ethnicity and Indigenous Americans/Alaskan Natives in the United States, as well as indigenous people in many other countries.<sup>27</sup>

### Social Determinants

Socioeconomic factors play an important role in CKD risk. People with lower socioeconomic status face greater burden of CKD and rapid GFR decline in the United States, and this finding has been replicated in many other developed countries.<sup>4,7</sup> Socioeconomic status is often lower in racial and ethnic minorities, and several studies have demonstrated attenuation or elimination of racial disparities in kidney outcomes with accounting for differences in income and the presence of health insurance, including risk for AKI, eGFR decline, and delay in transplantation in children.<sup>28</sup>

### Comorbid Conditions

Diabetes mellitus and hypertension are often cited as the major attributable causes of CKD in the developed world, and recent studies suggest similarities in the developing world.<sup>4</sup> Nearly half of the population with kidney failure has diabetes. Persons with diabetes and CKD merit concern, not only for the risk for albuminuria and CKD progression, but also for heightened risk for many other adverse outcomes, particularly cardiovascular disease. Risk for CKD among those with diabetes mellitus may be managed through tight control of blood glucose and blood pressure and the use of specific therapies such as angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and sodium-glucose cotransporter-2 inhibitors. Hypertension also has shown a graded association with the risk for KRT in observational studies, although the relationship with kidney disease is more complex than that of diabetes and CKD because hypertension is thought to be both a cause and a consequence of CKD. Obesity may confer increased risk for CKD over the long term, possibly mediated through the development of diabetes and hypertension, and smoking has also been linked to higher KRT risk.<sup>29</sup>

## ISSUES WITH IDENTIFYING CHRONIC KIDNEY DISEASE

### Lack of Awareness

Generally, a silent disease until the most advanced stages, CKD is largely underrecognized by providers and patients. Fewer than 10% of people with laboratory-confirmed CKD in the NHANES survey reported being aware that they had kidney disease, a finding corroborated in other regions.<sup>4</sup> Provider awareness may not be much higher; only 12% of patients with CKD had an associated diagnostic code in Stockholm health care.<sup>30</sup> Low awareness of disease may lead to low testing and referrals, with only 27% of patients with eGFR less than 60 mL/min/1.73 m<sup>2</sup> tested for albuminuria and 23% referred to a nephrologist. Albuminuria testing is also low in persons with diabetes. In the Stockholm cohort, 38% of patients with diabetes were evaluated for albuminuria, very similar to the testing frequency among US Medicare patients with diabetes.<sup>4</sup>

### Imperfect Biomarkers

Another issue in CKD identification is the occasional fallibility of existing biomarkers. Serum creatinine, the most widely used

filtration marker to estimate GFR, has non-GFR determinants, including diet and muscle mass. As a consequence, amputees and persons with reduced muscle mass may have a falsely elevated eGFR when creatinine is used. Thus, the KDIGO guidelines recommend a confirmatory measurement performed for persons in whom measuring creatinine may be less accurate.<sup>7</sup> To confirm diagnosis, KDIGO recommends obtaining a cystatin C–based eGFR or clearance measurement, or measuring GFR with an exogenous filtration marker when treatment decisions are at stake. Of note, cystatin C has its own non-GFR determinants, including body mass index and inflammation. For this reason, for the general population the most accurate estimating equation uses both creatinine and cystatin (see Chapter 3).<sup>31</sup>

### Errors in Urine Albumin Assessment

Variability in albumin excretion and measurement can confound CKD identification. In clinical practice, spot samples are usually assessed and standardized to urine creatinine to correct for error induced by differences in hydration status. However, urine albumin and creatinine excretion are thought to follow a circadian rhythm, additionally varying with posture, exercise, and diet.<sup>32</sup> Moreover, many providers instead quantify urine protein, a less sensitive and specific marker of glomerular permeability. Urine protein measurement is hampered by interlaboratory variation in methods and calibration and the lack of a reference measurement protocol.<sup>7</sup> For these reasons, urine albumin assessment is generally preferred over urine protein assessment, and repeated 24-hour collections offer the most accurate estimation of true albuminuria.

## OUTCOMES OF CHRONIC KIDNEY DISEASE

### Associations With Adverse Outcomes

The public health implications of CKD are enormous because of the strong associations with adverse outcomes—in particular, cardiovascular disease, kidney failure, and mortality. Some have argued that CKD should be considered a coronary heart disease risk equivalent, much like diabetes, and some guidelines recommend preventive statin use for all patients with CKD older than 50 years.<sup>33</sup> The independent associations of GFR and albuminuria with adverse outcomes has been demonstrated in general, high-risk, and CKD populations, as well as in separate geographic regions.<sup>8,34</sup>

### Variation in Absolute Risk for Adverse Outcomes

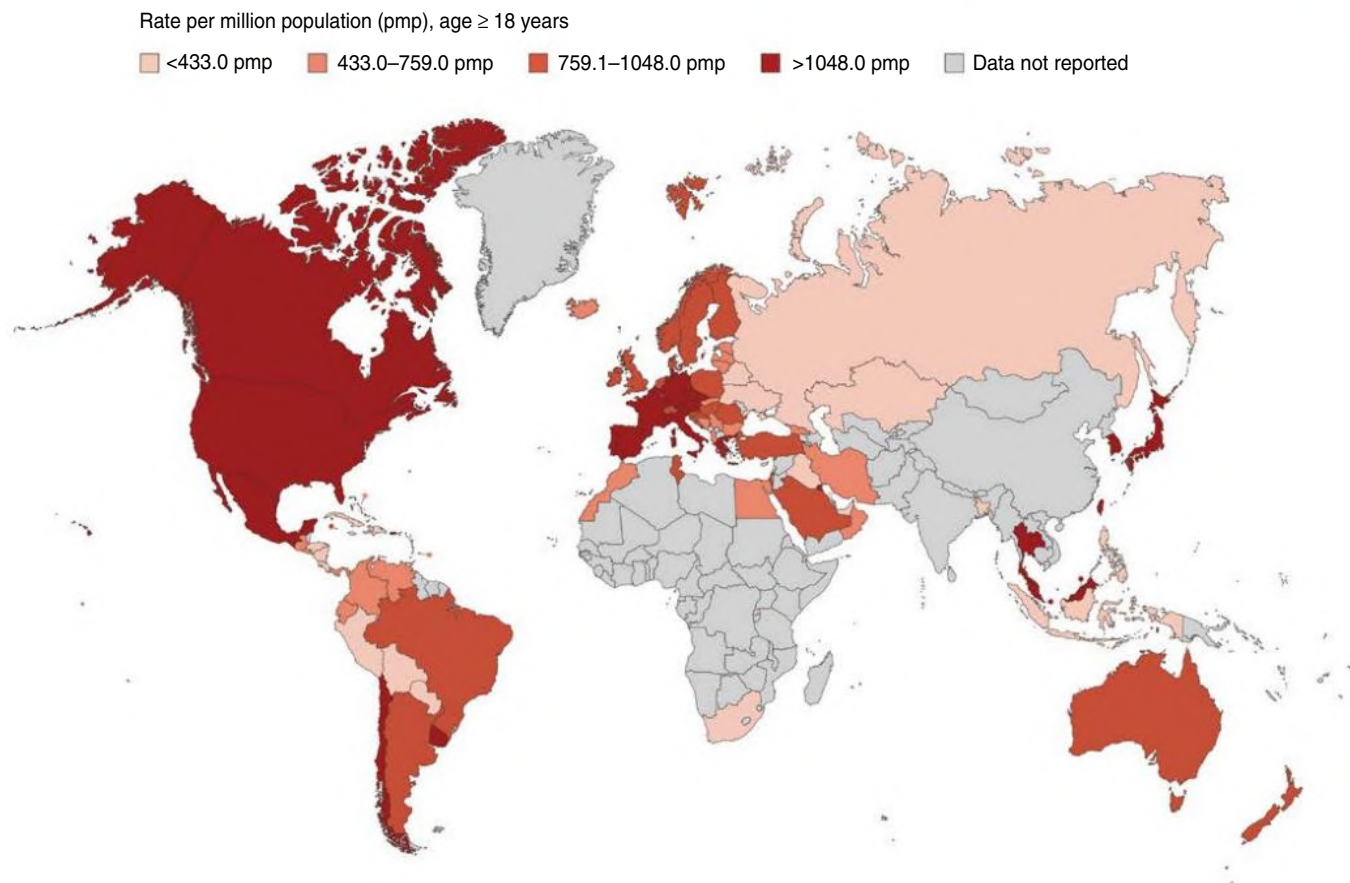
Increasingly, risk scores are used to inform patient counseling and treatment decisions. Guidelines encourage the incorporation of kidney failure risk prediction in kidney replacement therapy planning, and several risk scores now exist.<sup>7</sup> The kidney failure risk equation is the most widely used and has been globally validated.<sup>24</sup> The absolute risk for progression to kidney failure was observed to be slightly lower in non–North American countries, a difference that was not explained by differences in demographic or clinical characteristics. Variation in absolute risk for KRT thus might suggest differences in treatment availability, patient or provider acceptance, or differences in the competing risk for death.

## DIALYSIS EPIDEMIOLOGY

The number of people receiving chronic KRT (dialysis or kidney transplantation) worldwide has progressively increased over the 50 years since these treatments became available, and this increase continues. Initially available only to selected patients in restricted centers in some countries, both long-term dialysis and kidney transplantation



## Global Prevalence of Treated ESKD



**Fig. 80.5** Patients receiving kidney replacement therapy in 2010. ESKD, End-stage kidney disease. (From Bello A, et al. *Global Kidney Health Atlas: A Report by the International Society of Nephrology on the Global Burden of End-Stage Kidney Disease and Capacity for Kidney Replacement Therapy and Conservative Care Across Countries and Regions*. International Society of Nephrology; 2019.)

have become progressively more available not just in high-income countries but also in many low-income countries. Approximately 2.5 million people were estimated to be receiving chronic KRT in 2010<sup>35</sup>; absolute rates are high in North America, but the highest prevalent rates (people currently receiving KRT) per million population are in Taiwan and Japan (Fig. 80.5), where the vast majority of people receive long-term dialysis. Good data are available throughout the developed world because of the existence of regional and national dialysis registries; conversely, information such as this is less commonly available in lower-income countries.<sup>36</sup> These registries typically report rates, outcomes, and practice patterns of dialysis at a national level; many also have a major role in provision of hospital-specific information and other safety and quality reporting, as well as providing a substantial resource for clinical research.

### Incidence and Prevalence of Chronic Dialysis

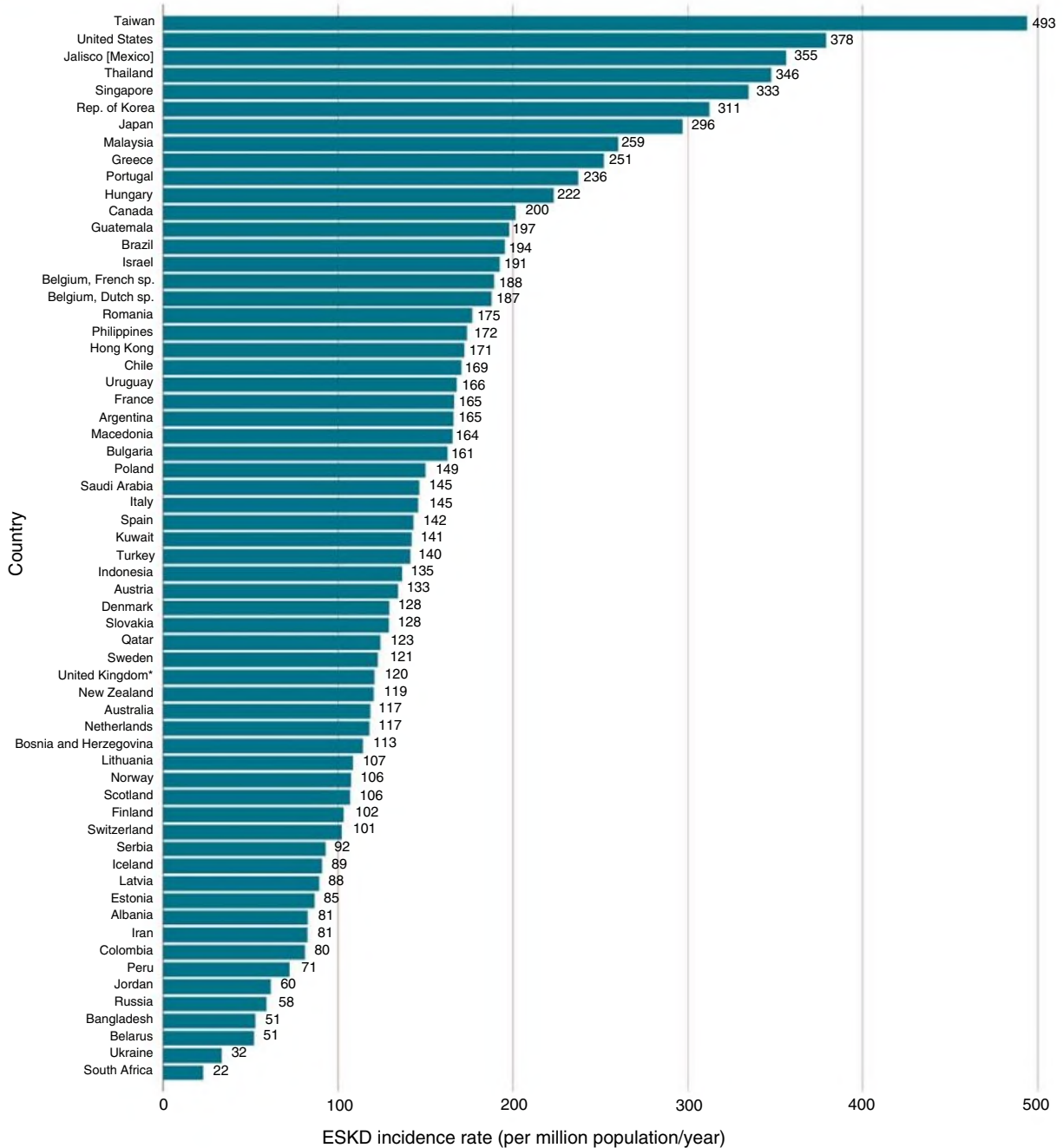
Incidence rates (the number of new KRT patients) comprise patients starting chronic or long-term dialysis and a small number of people receiving preemptive kidney transplants. Incidence rates are highest in Taiwan, Mexico, and the United States, followed by various other Asian countries. There is substantial variation among countries in every region around the world (Fig. 80.6). Diabetic kidney disease (DKD) has been the main factor in the increase in incidence rates in

almost all countries. In most countries, DKD is attributed as the cause for kidney failure in 40% to 60% of cases.

The trends in incidence rates vary across different regions. In many Asian countries (e.g., Korea, Singapore, Malaysia, Thailand), rates continue to increase. However, in the United States, Canada, Western Europe, Australia, and New Zealand, incidence rates have generally stabilized over the last 5 to 10 years. Nevertheless, in these countries the *prevalent* numbers and rates of dialysis continue to increase because the number of people leaving dialysis programs (deaths and transplant recipients) is smaller than the incident number.

Not every person with kidney failure receives dialysis or transplantation. In some situations (particularly those with substantial comorbidities or limited life expectancy), dialysis may not be clinically appropriate, or a decision is made for a conservative (nondialysis) approach to treatment (see Chapter 95). Registries typically report rates of KRT, which is synonymous with ESKD, a US administrative term meaning treatment with dialysis or kidney transplantation for chronic kidney failure. Estimating the “total burden” of kidney failure requires incorporation of both the number of people commencing KRT and the number of people dying of kidney failure without treatment. This has been examined using data linkage (of registry data with death certificates) in Australia<sup>37</sup> and in cohort studies, most notably in Alberta, Canada.<sup>37</sup> These studies suggest that among older age groups (over 75 years), dialysis therapy is used in

## Incidence Rate of Kidney Replacement Therapy by Country, 2014



**Fig. 80.6** Incidence rate of kidney replacement therapy (per million population per year), by country for 2014. \*Includes England, Wales, and Northern Ireland. ESKD, End-stage kidney disease. (From US Renal Data System. *International Comparisons. USRDS 2018 Annual Data Report. Vol. 2. Atlas of End-Stage Renal Disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2018.)

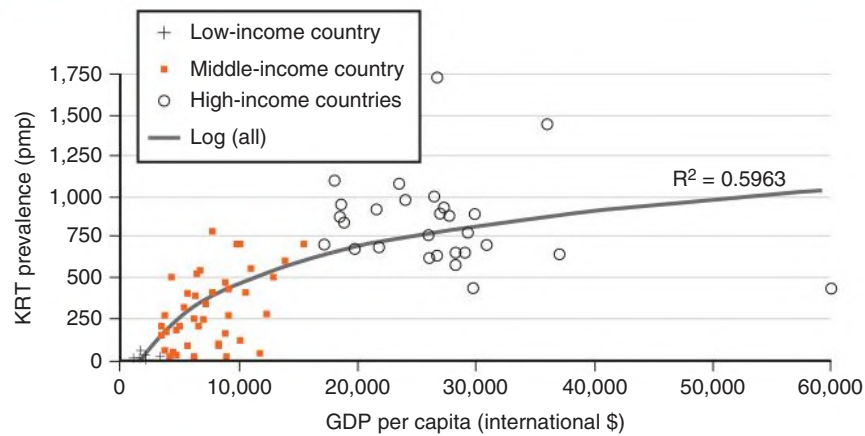
fewer than 50% of people with kidney failure. Consistent with this is evidence suggesting that among older patients with comorbidities, there is little or no survival advantage of dialysis therapy.<sup>38</sup>

Dialysis is a resource-intensive therapy, and there is a relationship between the provision of (and access to) dialysis and the overall economic development of a country (Fig. 80.7). As with other chronic diseases, there is also a gradation seen within many societies in rates of

incident kidney failure, with higher rates seen among those at the lower end of the socioeconomic spectrum.<sup>39,40</sup> Gradients with income and education are also seen among predialysis patients.<sup>41</sup>

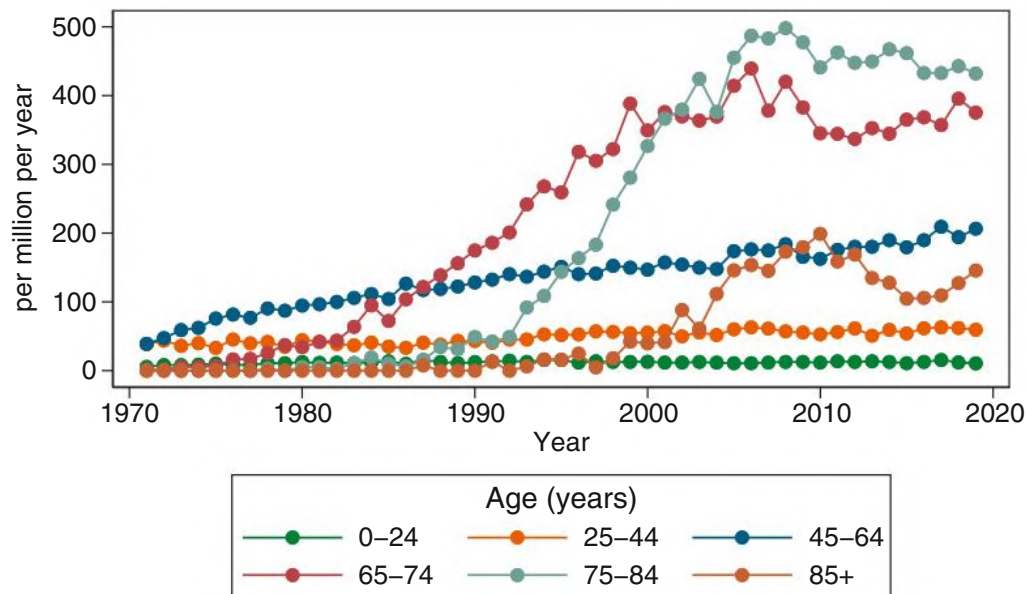
Incidence rates vary with sex, with rates generally lower among women than men.<sup>42</sup> Incidence rates increase with older age, up to 75 to 80 years old. There have been progressive changes differing across age-specific incidence rates over many years, with increasing incidence

### Prevalence of Patients Receiving Kidney Replacement Therapy as a Function of Per Capita Gross Domestic Product



**Fig. 80.7** Prevalence of patients receiving kidney replacement therapy (KRT) as of December 31, 2002, and gross domestic product (GDP) per capita. *PMP*, Per million population. (From White SL, Chadban SJ, Jan S, et al. How can we achieve global equity in provision of renal replacement therapy? *Bull World Health Organ.* 2008;86:229–237.)

### Age-Specific Incident KRT Rates in Australia



**Fig. 80.8** Age-specific kidney replacement therapy (KRT) incidence rates in Australia. (From ANZDATA Registry: 43rd Annual Report, 2020.)

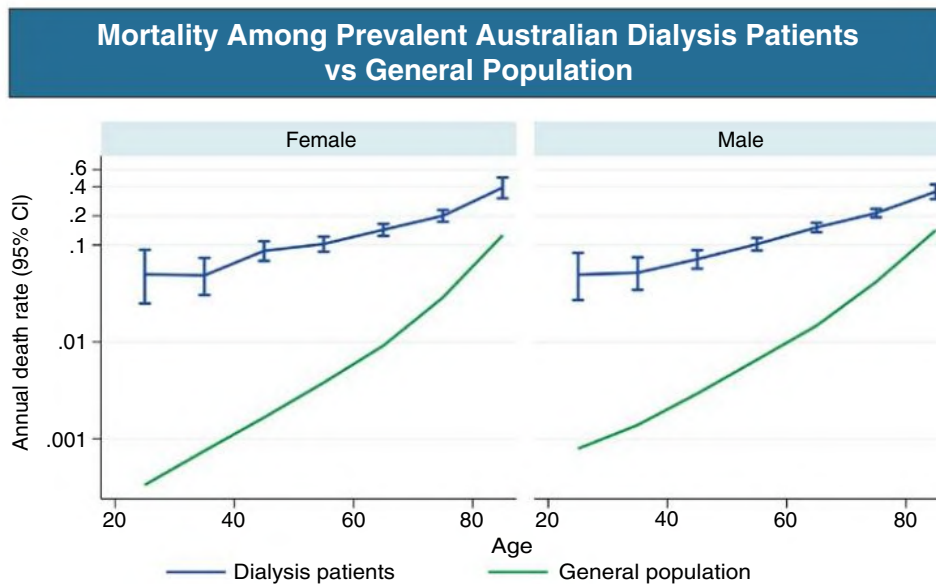
rates of progressively older age groups over a long period followed by stabilization in more recent years. An example of this (from Australia, but typical of that seen in high-income countries) is shown in Fig. 80.8.

Some ethnic groups experience particularly high rates of kidney failure and dialysis. Indigenous peoples in Australia, New Zealand, Canada, and the United States have very high rates of kidney disease, poorer outcomes, and less access to transplantation.<sup>43</sup> The causes underlying this are complex, but common elements appear to include

very high rates of diabetes, metabolic syndrome, and hypertension driving high rates of CKD.

### Dialysis Practice Patterns

Dialysis practice patterns vary widely across countries, with different usage of PD, HD, and home HD. For prevalent patients, the regions with the highest proportion of PD are Hong Kong (72% of prevalent patients) and the Jalisco region of Mexico (47% of prevalent patients).<sup>44</sup>



**Fig. 80.9** Death rates of prevalent dialysis patients versus general population. *CI*, Confidence interval. (From ANZDATA Registry: 43rd Annual Report, 2020.)

Home HD is most commonly used in New Zealand and Australia, where 18% and 9%, respectively, of prevalent KRT patients are treated with this modality. Practice patterns also vary in the application of various techniques of HD; for example, dialysis sessions are generally longer in Japan, with slower blood flow rates.

## Outcomes

The rationale for dialysis is to prevent death from uremia, and hence the outcome for dialysis treatment is usually reported as mortality rate. Other important outcomes commonly reported include cause-specific mortality rates, hospitalizations, and technique failure. Patient-reported outcomes (including measures of quality of life and functional ability) are also extremely important.

Crude mortality rates vary widely; a multinational observational study (Dialysis Outcomes and Practice Patterns Study) reported rates from 6.6% per year in Japan to 21.7% in the United States.<sup>45</sup> Mortality rates among dialysis patients are substantially higher than in the general population, a differential particularly marked among younger age groups (Fig. 80.9). The excess mortality is primarily driven by cardiovascular and infectious causes. There is also a modest (approximately 3-fold) excess risk for cancer among dialysis patients, even after exclusion of cancers directly related to the cause of kidney disease (see Chapter 92).<sup>46</sup> An issue unique to categorization of deaths among the dialysis cohort is assignment of deaths as a result of withdrawal from therapy. There is a wide range in reported frequency of withdrawal, from less than 5% to more than 20% of all deaths.<sup>47</sup> These differences may in part reflect different coding mechanisms but are also likely to reflect differences in practice patterns in the commencement of therapy among older patients with higher levels of comorbidity.

There is variation in mortality over time, with the highest rates seen in the first 3 months after commencing dialysis, particularly among older age groups (Fig. 80.10). After this period, mortality rates approximate the longer-term rates. There are a number of possible contributors to this, including the risk associated with commencement of dialysis (such as the placement of central venous catheters) and increased severity of comorbidities associated (or causing) the deterioration of kidney function.

Multiple observational studies have compared mortality between PD and HD, but no definitive randomized controlled trials (RCTs) have been completed despite several attempts. Results from observational

studies have varied across countries; common elements are the presence of multiple interactions such that the comparison depends on comorbidities and patient age, the length of follow-up, and selection biases and local/regional outcomes of each technique.

For PD (and to a lesser extent home HD), “technique survival” is a common metric. This refers to the proportion of people continuing on PD at various time points. It is a combined end point, with patient death and transfer to HD considered failures. Typically, 30 days of HD is considered to be a permanent transfer to HD. It is important to ensure consistency of definitions in comparisons of technique survival; alternative duration definitions have substantial effects on the calculated technique survival.

The use of patient-reported outcomes (PROMs) is growing, supported by evidence that patient priorities around dialysis treatment differ from those of clinicians.<sup>48</sup> However, there are significant limitations to the uptake of PROMs. First, the resources required for implementation of PROMs in routine practice can be a substantial barrier.<sup>49</sup> Second, despite the conceptual simplicity, there is limited evidence to date about how measuring PROMs may affect quality of life.

## EPIDEMIOLOGIC CONCEPTS

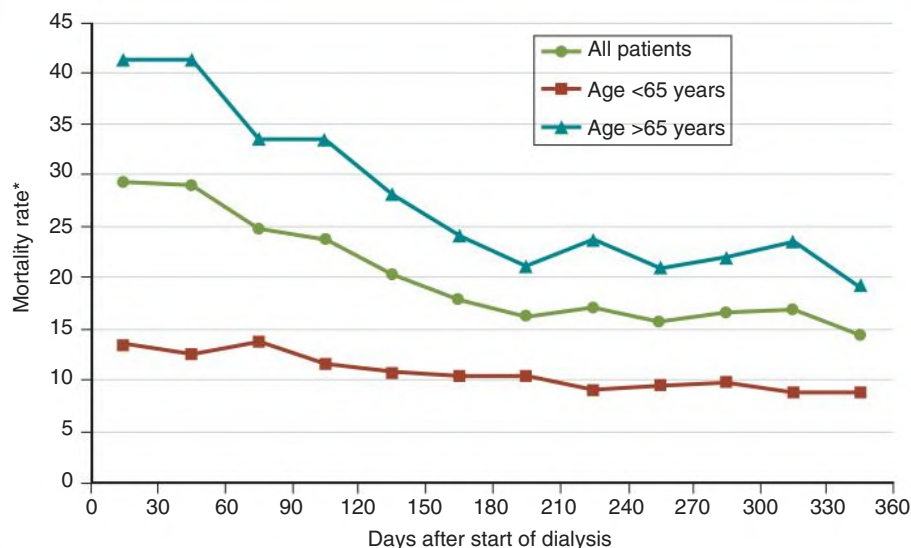
Epidemiologic factors influence the reported rates of KRT; understanding the potential biases is a necessary part of interpretation of these data.

The *prevalent* number of KRT patients at any point in time receiving a form of KRT reflects both the rate of people starting treatment (for overall KRT numbers, the *incidence* rate) and the rate of people ceasing treatment (the *mortality* rate). These rates can be considered *transition probabilities* among various states. This concept can be applied to the overall number of people receiving KRT, the number receiving dialysis or transplantation, and other subgroups. For example, the determinants of the prevalent number of people receiving dialysis at any point in time can be determined from the existing population plus the number of new patients (incidence rate plus the number of people who have lost graft function) less the number of deaths (mortality rates during dialysis) and kidney transplants.

This form of multistate model is often known as a Markov model. Use of these models to estimate future trends can be used by health



## Mortality Rates by Age Among Hemodialysis Patients



| Mortality rate during the first 360 days of HD treatment |      |       |       |        |         |         |         |         |         |         |         |         |
|----------------------------------------------------------|------|-------|-------|--------|---------|---------|---------|---------|---------|---------|---------|---------|
|                                                          | 0-30 | 31-60 | 61-90 | 91-120 | 121-150 | 151-180 | 181-210 | 211-240 | 241-270 | 271-300 | 301-330 | 331-360 |
| All patients                                             | 29.3 | 28.9  | 24.9  | 23.8   | 20.4    | 17.9    | 16.2    | 17.0    | 15.7    | 16.5    | 16.8    | 14.4    |
| Age <65 years                                            | 13.3 | 12.6  | 13.5  | 11.6   | 10.7    | 10.4    | 10.2    | 9.1     | 9.3     | 9.7     | 8.9     | 8.7     |
| Age >65 years                                            | 41.2 | 41.2  | 33.7  | 33.5   | 28.2    | 24.0    | 21.1    | 23.6    | 21.0    | 21.9    | 23.5    | 19.2    |

**Fig. 80.10** Overall mortality rates among hemodialysis patients by age. \*Number of deaths per 100 patient-years. HD, Hemodialysis. (From Robinson BM, Zhang J, Morgenstern H, et al. Worldwide, mortality risk is high soon after initiation of hemodialysis. *Kidney Int.* 2014;85:158–165.)

services and others in predicting the future number of people receiving KRT. This can be done under existing conditions and under various hypothetical scenarios to model the effects of various interventions.

The incidence rates reported for ESKD typically exclude dialysis for AKI. Definitions of the start point of KRT vary across registries and data collections. In many cases, an “intent” definition is used (i.e., dialysis is commenced with the intention of long-term treatment, underpinned by a diagnosis of irreversible kidney failure). The other approach (used by the US registry and some other registries) is to use day 90 after dialysis start as the point at which patients are deemed to require “chronic” dialysis. Although this approach eliminates uncertainty around people with prolonged acute (or acute on chronic) kidney injury who may or may not recover, it does prevent analysis of early mortality.

**Selection bias** affects incidence rates. There may be constrained provision of dialysis services that affects ability to access dialysis services. There have also been changes in clinical practice over time affecting the propensity to treat in certain situations (especially among older and more frail patients). These factors will influence observed rates; differences in these selection biases will be relevant to comparisons of disease incidence among groups and also to the same group between different times. This is likely to have contributed to the increasing dialysis rates among progressively older people observed in the 1970s to late 1990s.

**Lead time bias** occurs when the time point of observation differs between groups. One example of this is the comparison of PD and HD cohorts. Patients commencing KRT with PD typically do so at a higher eGFR than those commencing with HD; current UK data suggest this mean difference is around 0.5 mL/min/1.73 m<sup>2</sup>.<sup>50</sup> Although small, this may amount to 2 to 3 months of difference in starting point of analyses.

**Competing risks** arise when an illness may lead to multiple possible outcomes, but the occurrence of one precludes another. For example,

changes in cardiovascular mortality may affect the incidence rate of KRT because patients who survive despite significant heart disease may avoid premature death that would preclude dialysis treatment. Another example is the influence of kidney transplantation in evaluation of survival of a dialysis cohort: increased occurrence of transplantation may lead to higher mortality in a dialysis cohort.

**Immortal time bias** arises when an inappropriate time point is chosen for comparators. An example of this might be comparison of mortality after coronary angiography performed as a pretransplant assessment between those transplanted and not transplanted; the transplanted group will accumulate “immortal” follow-up time between angiography and transplantation, in contrast to the nontransplanted group.

## EVIDENCE QUALITY

There is a relative dearth of RCTs supporting practice in the area of dialysis compared with other disciplines. Most large-scale trials examining dialysis-specific interventions have yielded negative or inconclusive results; for example, the use of non-calcium-containing phosphate binders,<sup>51</sup> increased dialysis dose,<sup>52</sup> use of statins in dialysis patients,<sup>53</sup> and hemodiafiltration.<sup>54</sup> In response to this, clinical trials networks with a focus on nephrology have been established in several countries to catalyze the conduct of trials (particularly where there is limited commercial interest). Examples of this approach currently underway are multicenter randomized trials examining issues such as dialysate sodium concentration and the effect of differing phosphate targets.

The outcomes examined in trials also have been critically examined. Due to the time delay and multiple contributing factors to mortality, many trials have used surrogate outcomes. Given the association of many biochemical abnormalities with increased mortality, their use

appears logical. However, as in other areas of medicine, use of surrogate outcomes in trials and evidence generation in nephrology has been problematic, with changes in surrogates not reflected by reduction in mortality or other hard endpoints. An example of this is the use of phosphate (and parathyroid hormone) as a surrogate to assess

the effectiveness of cinacalcet. Partly as a response to this, initiatives have been instituted to develop and standardize the outcomes for trials.<sup>55</sup> An important element of this is also the inclusion of input from patients and consumers of services in setting research priorities.

## SELF-ASSESSMENT QUESTIONS

1. The KDIGO CKD classification includes all of the following in the staging of CKD *except*:
  - A. the severity of hypertension.
  - B. the severity of albuminuria.
  - C. the level of GFR.
  - D. the cause of disease.
2. Which statement is *true*?
  - A. Low GFR is unimportant in older adults.
  - B. Higher albuminuria is worse for men than women.
  - C. Higher albuminuria is associated with acute kidney injury.
  - D. Assays for proteinuria are more accurate than assays for albuminuria.
3. Racial disparities in kidney failure incidence are in part attributable to:
  - A. the higher prevalence of *APOL1* genetic risk variants in those of European descent.
  - B. lower socioeconomic status and poorer access to health care among Black individuals.
  - C. greater rates of smoking among Black individuals.
  - D. lipoxins.
4. Which statement is *most true* about rates of new dialysis patients?
  - A. They are highest in Eastern European countries.
  - B. An increase in rates of glomerulonephritis is the major driver.
  - C. Rates are highest among young people.
  - D. Rates are higher among men than women.
  - E. Rates are generally lower among groups with lower socioeconomic status.
5. Which statement is *true* regarding competing risks in comparison of rates of new dialysis patients between groups?
  - A. Competing risks refers to higher rates seen among health systems in which funding is allocated on a competitive grants basis.
  - B. Men more than women are affected because testosterone leads to greater competitive drive.
  - C. Competing risks from transplantation will not be a factor if measured covariates are included and adjustment is made for them.
  - D. Incident dialysis rates might be higher if vascular mortality associated with CKD were lower.
  - E. Propensity score matching can largely address the problems raised by competing risks.

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